EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Preparation of Grignard reagent & Acid Base Reaction

1. Grignard reagent can be prepared by

(A) $CH_3—CH_2—Cl + Mg \xrightarrow[\text{ether}]{\text{dry}}$

(B) $CH_3—\underset{Cl}{CH}—\underset{OH}{CH_2} + Mg \xrightarrow[\text{ether}]{\text{dry}}$

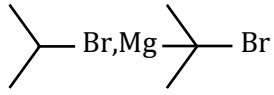
(C) $CH_3—\overset{CH_3}{\underset{CH_3}{C}}—OH + Mg \longrightarrow$

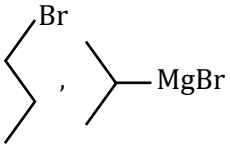
(D) All of them

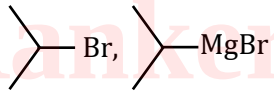
CGR060

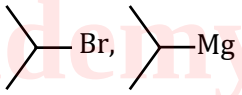
2. $X + Mg \xrightarrow{\text{dry ether}} Y \xrightarrow{D_3O^{\oplus}} CH_3\underset{D}{CH}CH_3$

Identify X and Y –

(A) 

(B) 

(C) 

(D) 

CGR061

3. The correct order of reactivity of halides with Mg for the formation of Grignard reagents :-

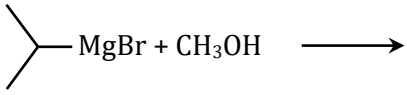
(A) RI > RCl > RBr (B) RBr > RCl > RI (C) RI > RBr > RCl (D) RCl > RBr > RI

CGR062

4. Propane is not formed when C_3H_7MgBr is treated with :-

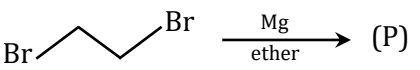
(A) H_2 (B) Phenol (C) Ethanoic acid (D) 1–Butyne

CGR063

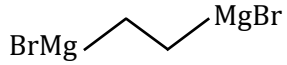
5.  , product (X) is:

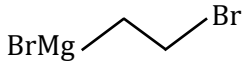
(A)Propane (B) Ethane (C) Butane (D) Propan–2–ol

CGR064

6. 

(P) is :-

(A) 

(B) 

(C) $H_2C{=}CH_2$

(D) All

CGR065

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7. $C_2H_5MgBr + CH_3—NH_2 \longrightarrow (P)$
Product (P) is :-
(A) C_2H_6 (B) C_3H_8 (C) $C_2H_5NH_2$ (D) No reaction

CGR066

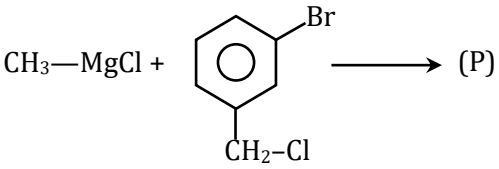
8. Which gas is released during the following reaction :-
$HC\equiv CH + MeMgBr \longrightarrow$
(A) Ethane (B) Propane (C) Methane (D) Butane

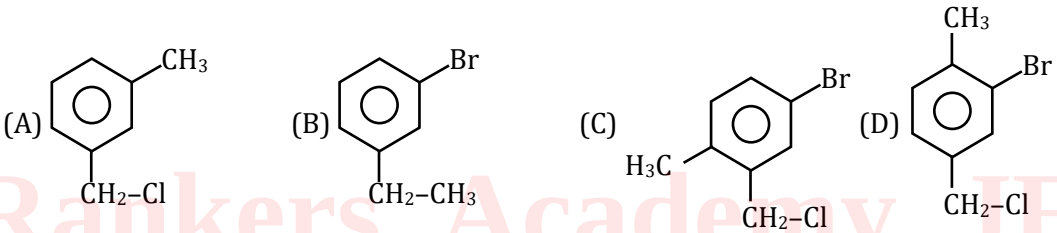
CGR067

9. Which of the following will undergo acid-base reaction with Grignard reagent :-
(A) $HC\equiv CH$ (B) R–OH (C) $R–CO_2H$ (D) All of these

CGR068

10. Find the major product for the given following reaction :



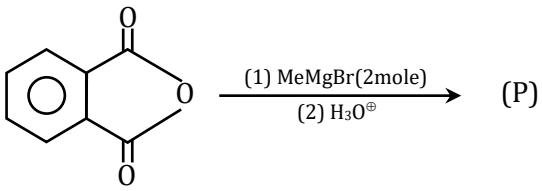


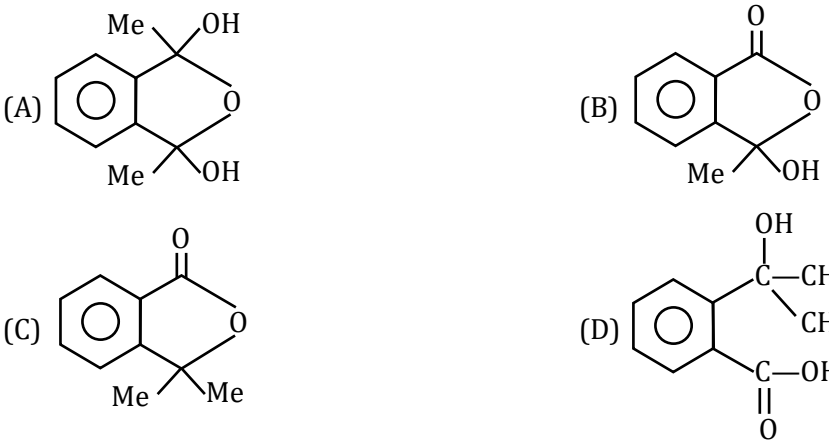
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CGR069

Chemical Reaction of Grignard Reagents (NAER, NAR, SN Reaction)

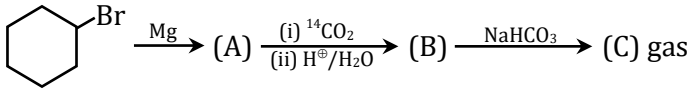
11. For the reaction the product expected is :





CGR070

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12. 

Product C is

(A) CO

(B) $^{14}CO_2$

(C) CO_2

(D) A mixture of $^{14}CO_2$ and CO_2

CGR073

13. $Br-CH_2C\equiv C-CH_2-Br \xrightarrow[Et_2O]{Mg\ (1mole)}$ Product.

The major product is ?

(A) $Br-Mg-CH_2-C\equiv C-CH_2-Br$

(B) Cyclobutyne

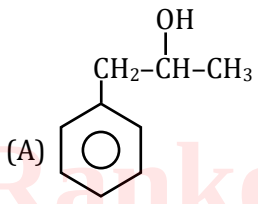
(C) $-\{CH_2-C\equiv C-CH_2\}-n$

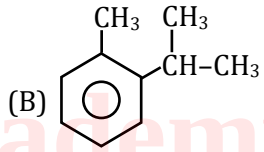
(D) $CH_2=C=C=CH_2$

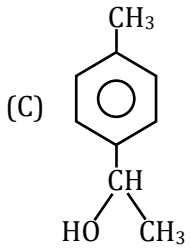
CGR071

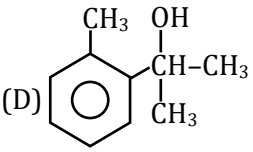
14. Find product C of the following reaction :

$PhCH_3 \xrightarrow[h\nu]{Cl_2}$ (A) $\xrightarrow[ether]{Mg}$ (B) $\xrightarrow[NH_4Cl]{CH_3CHO}$ (C)

(A) 

(B) 

(C) 

(D) 

CGR078

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

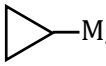
Reaction of Grignard's Reagent

1. From which compound Alcohol is formed when RMgX followed by $H_3O^{\oplus}$ is treated with :

(A) Ethanoyl chloride (B) O_2 (C) oxirane (D) trimethyl orthoformate

CGR079

2. In which of the following reactions 3° alcohol will be obtained as a product.

(A)  $\xrightarrow[H^{\oplus}]{}$

(B) $PhMgBr(excess) + CH_3-CO-Cl \xrightarrow[H^{\oplus}]{}$

(C) $CH_3MgBr(excess) + CH_3-CO-O-CO-CH_3 \xrightarrow[H^{\oplus}]{}$

(D) $CH_3MgBr(excess) + Cl-CO-OEt \xrightarrow[H^{\oplus}]{}$

CGR080

3. Select the correct statement :

(A) 1,4–dibromo butane react with excess of magnesium in ether to generate di–Grignard reagent.

(B) 1,2–dichlorocyclohexane treated with excess of Mg in ether produces cyclohexene.

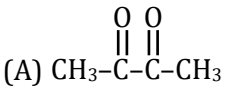
(C) Vicinal dihalides undergo dehalogenation to give alkene when heated with Zn dust or Mg.

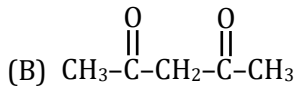
(D) 1,3–dichloropropane by treatment with Zn dust or Mg forms cyclopropane.

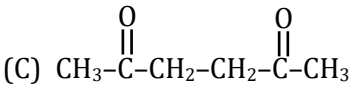
Bankers Academy JEE

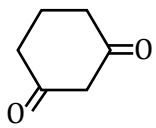
CGR081

4. Nucleophilic addition of Grignard reagent can not occur in

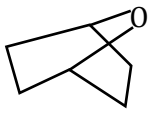
(A) 

(B) 

(C) 

(D) 

CGR082

5.  $\xrightarrow[(ii)\ H^{\oplus}/H_2O]{(i)\ CH_3MgBr}$ (X)

Correct statement about X is

(A) The product is optically active.

(B) The product contains plane of symmetry.

(C) The product shown geometrical isomerism.

(D) The product shown optical isomerism.

CGR083

For More Join: @IITJEE_Advanced

EXERCISE (S)

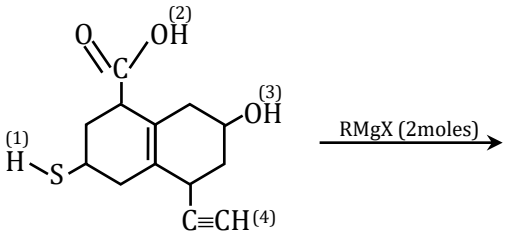
Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Preparation of Grignard reagent & Acid Base Reaction

1. How many total possible Grignard reagent can give 3–methylpentane on treatment with $H^{\oplus}$?

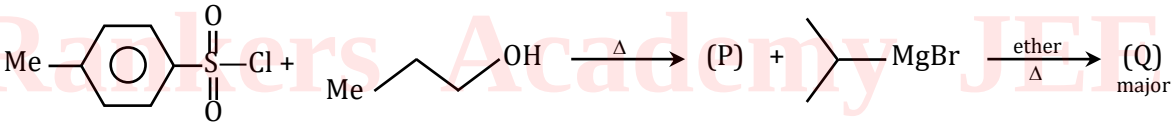
CGR084

2. 

The sum of the positions from where deprotonation will occur is

CGR076

Reaction of Grignard's Reagent

3. 

No. of chiral centre in product (Q).

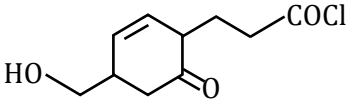
CGR074

4. Compounds are shown with the no. of RMgX required for complete reaction as (x), (y) and (z). what is x+y+z ?

(i) $CH_3COOC_2H_5$ (x) (ii) CH_3COCl (y) (iii) $HOCH_2COOC_2H_5$ (z)

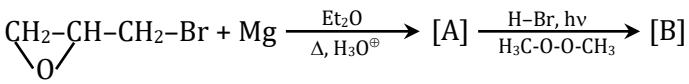
CGR075

5. Number of moles of RMgX required for complete reaction with one mole of the following compound is ?



CGR077

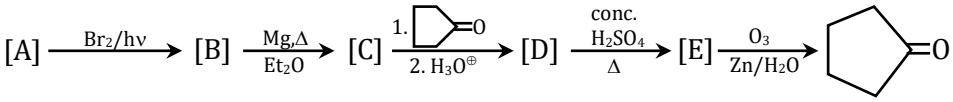
6. Choose the final product (B) of the following reaction.



What is the molecular weight of final product ?

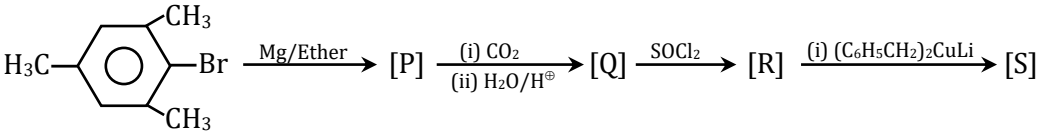
CGR127

For More Join: @IITJEE_Advanced

7. 

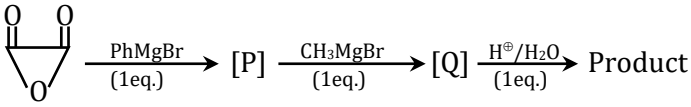
Find D.O.U. in compound [E] ?

CGR128

8. 

How many 2° carbon in product [S] ?

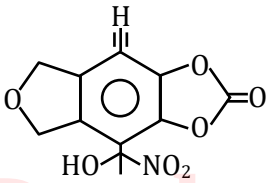
CGR129

9. 

Calculate total number of hydrogen atom present in final product ?

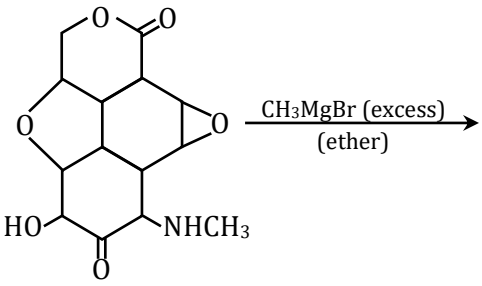
CGR130

10. Calculate the number of molecules of Grignard reagent consumed by 1 molecule of following compound.



CGR089

11. Find the number of CH_3MgBr molecules used the given reactant, to react completely.

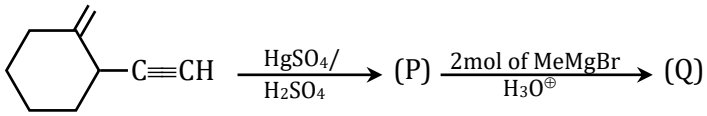


CGR090

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Reaction of Grignard's Reagent

12. 

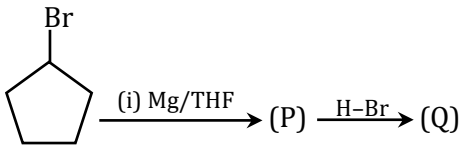
Number of 3° Alcohol formed in product Q

CGR026

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For More Join: @IITJEE_Advanced

13. How many carbonyl isomers of $C_5H_{10}O$ reacts with PhMgBr to give racemic mixture (excluding steroisomer).

CGR027

14. 

No. of sp^3 hybridized carbon in (Q) is :–

CGR028

15. How many moles of methane is formed when one mole of lactic acid reacts with excess methyl magnesium chloride ?

CGR029

16. The number of moles of Grignard reagent consumed per mole of the compound is

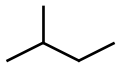
$CH_2(OH)–CH_2–CO–CH_2–CH_2–COEt$

CGR030

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For More Join: @IITJEE_Advanced

EXERCISE – JEE (Main) PYQ

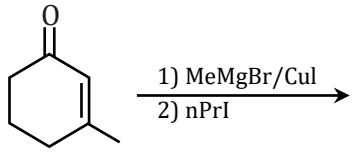
1. RBr $\xrightarrow[\text{(ii) } H_2O]{\text{(i) Mg, dry ether}}$  2-Methylbutane

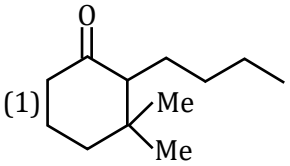
The maximum number of RBr producing 2-methylbutane by above sequence of reactions is ____.
(Consider the structural isomers only) **[JEE (Main) 2025]**

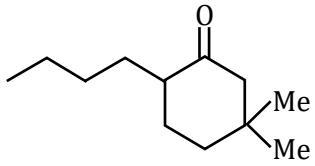
(1) 4 (2) 5 (3) 3 (4) 1

CGR132

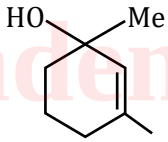
2. Find out the major product from the following reaction **[JEE (Main) 2023]**



(1) 

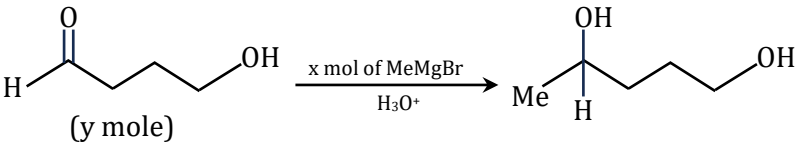
(2) 

(3) 

(4) 

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CGR107

3. 

The ratio x/y on completion of the above reaction is _____. **[JEE (Main) 2023]**

CGR108

4. $H_3C—CH_2—\underset{\underset{OH}{|}}{CH}—CH_3 \xrightarrow[\text{(iii) } D_2O]{\text{(i) NaI, } H_3PO_4,\ \text{(ii) Mg, Dry ether}} \underset{\text{Product}}{[X]}$

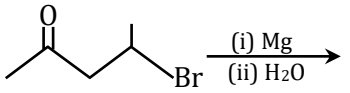
Product [X] formed in the above reaction is : **[JEE (Main) 2023]**

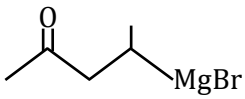
(1) $H_3C–CH_2–\underset{\underset{D}{|}}{CH}–CH_3$ (2) $H_3C–CH_2–\overset{\overset{H}{|}}{\underset{\underset{OH}{|}}{C}}–CH_3$

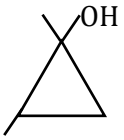
(3) $H_3C–CH_2–CH=CH_2$ (4) $H_3C–CH=CH–CH_3$

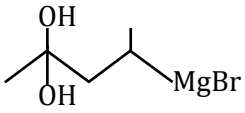
CGR109

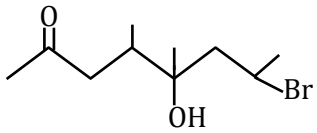
For More Join: @IITJEE_Advanced

5.  'A' (Major Product) A is **[JEE (Main) 2023]**

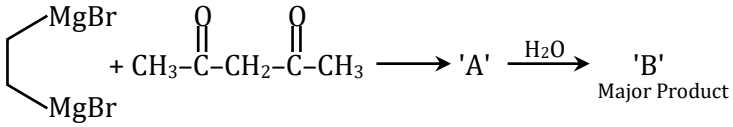
(1) 

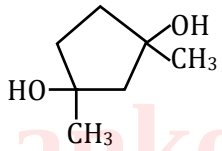
(2) 

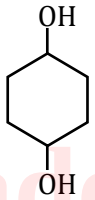
(3) 

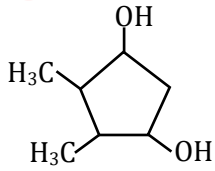
(4) 

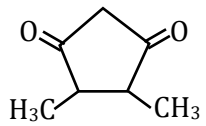
CGR110

6.  Consider the above reaction sequence and identify the product B. **[JEE (Main) 2022]**

(1) 

(2) 

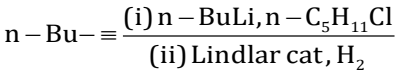
(3) 

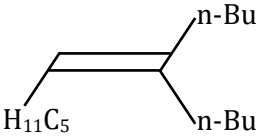
(4) 

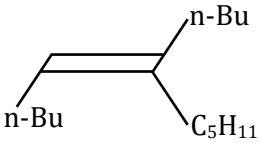
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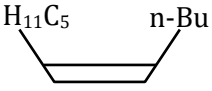
CGR100

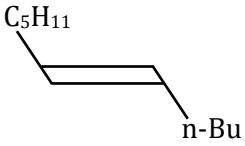
7. What will be the major product of following sequence of reactions? **[JEE (Main) 2022]**



(1) 

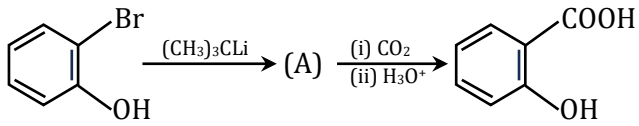
(2) 

(3) 

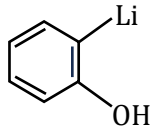
(4) 

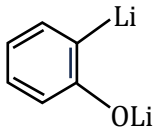
CGR101

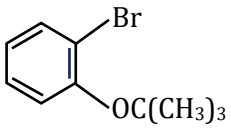
For More Join: @IITJEE_Advanced

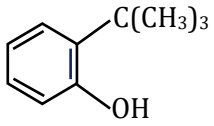
8. 

In the given conversion the compound A is: **[JEE (Main) 2022]**

(1) 

(2) 

(3) 

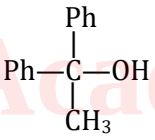
(4) 

CGR102

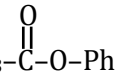
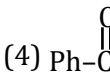
9. A sample of 4.5 mg of an unknown monohydric alcohol, R–OH was added to methylmagnesium iodide. A gas is evolved and is collected and its volume measured to be 3.1 mL. The molecular weight of the unknown alcohol is ___ g/mol. [Nearest integer] **[JEE (Main) 2022]**

CGR103

10. Which reactant will give the following alcohol on reaction with one mole of phenyl magnesium bromide (PhMgBr) followed by acidic hydrolysis ? **[JEE (Main) 2022]**



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(1) $CH_3-C\equiv N$ (2) $Ph-C\equiv N$ (3)  (4) 

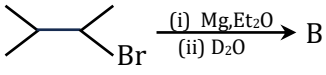
CGR104

11. A 100 mL solution of CH_3CH_2MgBr on treatment with methanol produces 2.24 mL of a gas at STP. The weight of gas produced is _____ mg. [nearest integer] **[JEE (Main) 2022]**

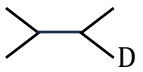
CGR105

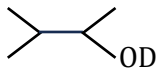
12. Compound 'A' undergoes following sequence of reactions to give compound 'B'. The correct structure and chirality of compound 'B' is: **[JEE (Main) 2022]**

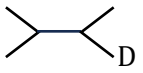
[where Et is $-C_2H_5$]

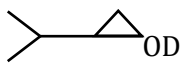


Compound 'A'

(1)  , Achiral

(2)  , Chiral

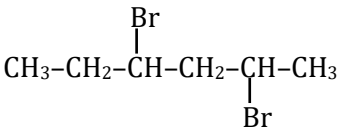
(3)  , Chiral

(4)  , Achiral

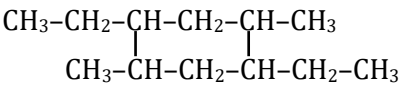
CGR106

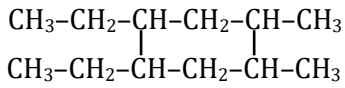
For More Join: @IITJEE_Advanced

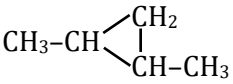
13. The product formed in the first step of the reaction of [JEE (Main) 2021]

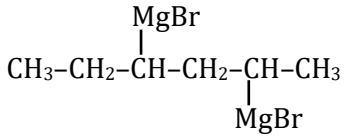


with excess $Mg/Et_2O(Et=C_2H_5)$ is:

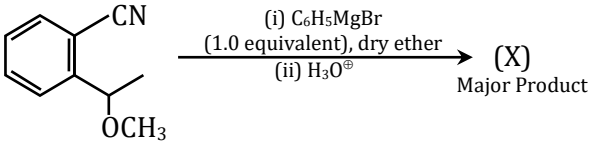
(1) 

(2) 

(3) 

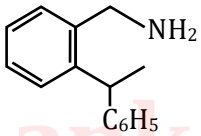
(4) 

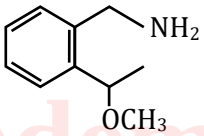
CGR035

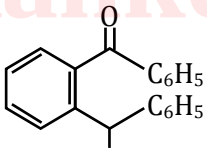
14. 

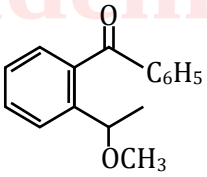
[JEE (Main) 2021]

The structure of X is :

(1) 

(2) 

(3) 

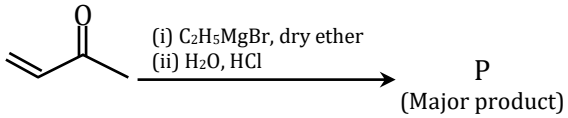
(4) 

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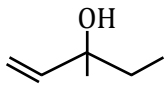
CGR036

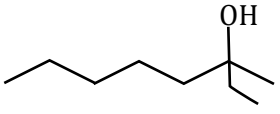
15. To synthesise 1.0 mole of 2-methylpropan-2-ol from Ethylethanoate _____ equivalents of CH_3MgBr reagent will be required. [JEE (Main) 2021]

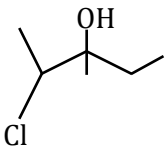
CGR037

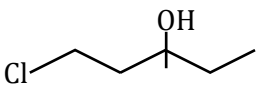
16. 

Consider the above reaction, the major product 'P' is: [JEE (Main) 2021]

(1) 

(2) 

(3) 

(4) 

CGR095

For More Join: @IITJEE_Advanced

17. The correct sequential addition of reagents in the preparation of 3-nitrobenzoic acid from benzene is: **[JEE (Main) 2021]**

(1) $Br_2/AlBr_3$, HNO_3/H_2SO_4, Mg/ ether, CO_2, H_3O^+

(2) $Br_2/AlBr_3$, NaCN, H_3O^+, HNO_3/H_2SO_4

(3) $Br_2/AlBr_3$, HNO_3/H_2SO_4, NaCN, H_3O^+

(4) HNO_3/H_2SO_4, Br_2/ $AlBr_3$, Mg/ether, CO_2, H_3O^+

CGR096

18. Given below are two statements : **[JEE (Main) 2021]**

Statement I : Ethyl pent–4–yn–oate on reaction with CH_3MgBr gives a 3°–alcohol.

Statement II : In this reaction one mole of ethyl pent–4–yn–oate utilizes two moles of CH_3MgBr.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement I** and **Statement II** are false.

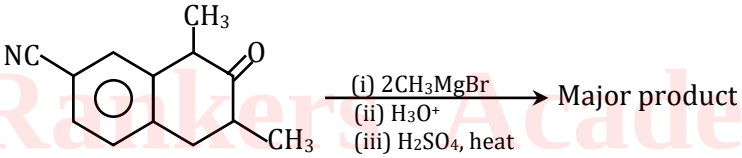
(2) **Statement I** is false but **Statement II** is true.

(3) **Statement I** is true but **Statement II** is false.

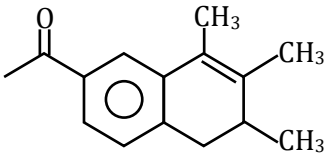
(4) Both **Statement I** and **Statement II** are true.

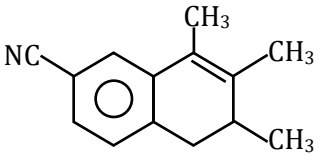
CGR039

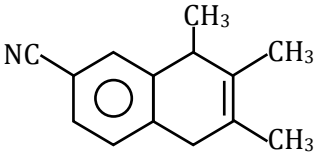
19. Which one of the following is the major product of the given reaction? **[JEE (Main) 2021]**

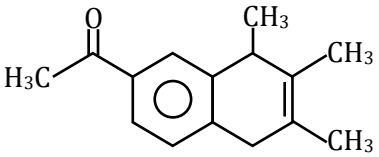


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(1) 

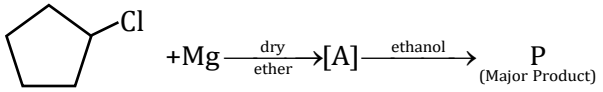
(2) 

(3) 

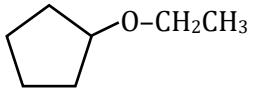
(4) 

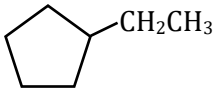
CGR097

20. In the following sequence of reactions the P is : **[JEE (Main) 2021]**



(1) 

(2) 

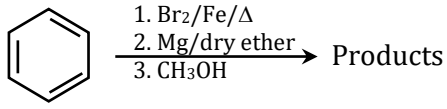
(3) 

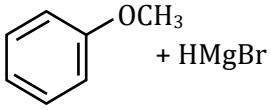
(4) 

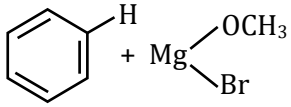
CGR038

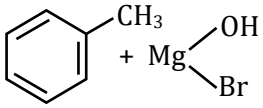
For More Join: @IITJEE_Advanced

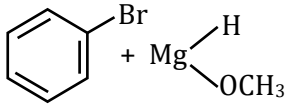
21. For the following : **[JEE (Main) 2021]**



(1) 

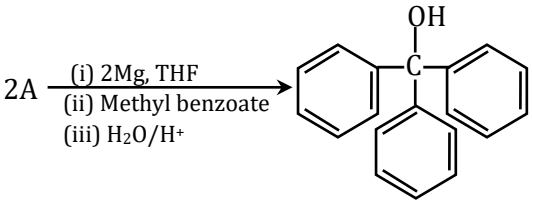
(2) 

(3) 

(4) 

CGR098

22. In the given reaction **[JEE (Main) 2021]**



'A' can be

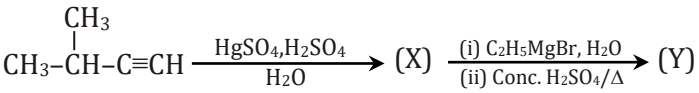
(1) benzyl bromide (2) bromobenzene

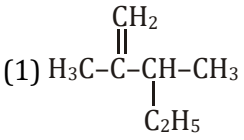
(3) cyclohexyl bromide (4) methyl bromide

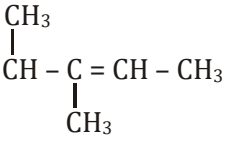
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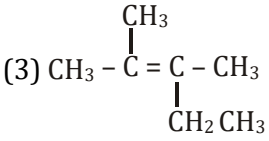
CGR099

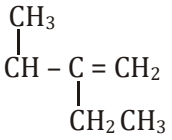
23. The major product (Y) in the following reactions is : **[JEE (Main) 2020]**



(1) 

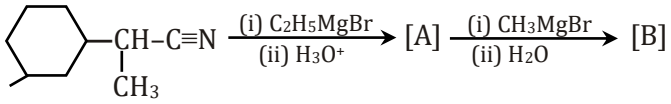
(2) 

(3) 

(4) 

CGR092

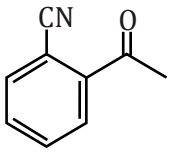
24. The number of chiral centres present in [B] is _____. **[JEE (Main) 2020]**

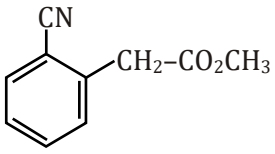


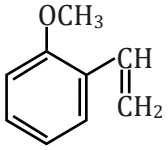
CGR093

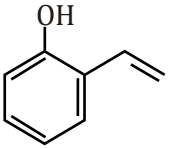
For More Join: @IITJEE_Advanced

25. Which of the following compounds reacts with ethylmagnesium bromide and also decolourizes bromine water solution :- **[JEE (Main) 2019]**

(1) 

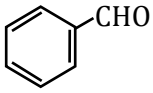
(2) 

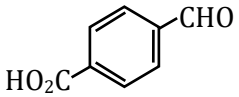
(3) 

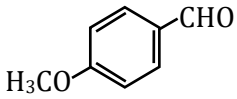
(4) 

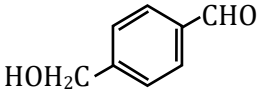
CGR091

26. The aldehydes which will not form Grignard product with one equivalent Grignard reagents are : **[JEE (Main) 2019]**

(A) 

(B) 

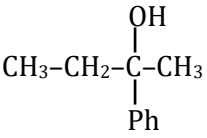
(C) 

(D) 

(1) (B), (C), (D) (2) (B), (D) (3) (B), (C) (4) (C), (D)

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CGR033

27.  cannot be prepared by : **[JEE (Main) 2019]**

(1) $HCHO + PhCH(CH_3)CH_2MgX$

(2) $PhCOCH_2CH_3 + CH_3MgX$

(3) $PhCOCH_3 + CH_3CH_2MgX$

(4) $CH_3CH_2COCH_3 + PhMgX$

CGR034

For More Join: @IITJEE_Advanced

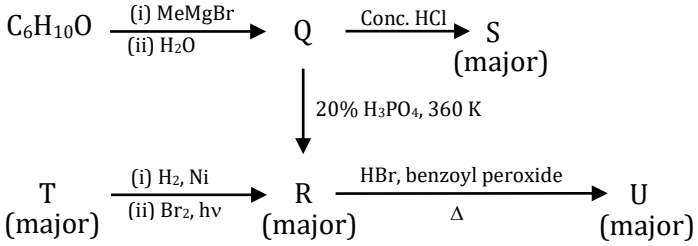
EXERCISE - JEE (Advanced) PYQ

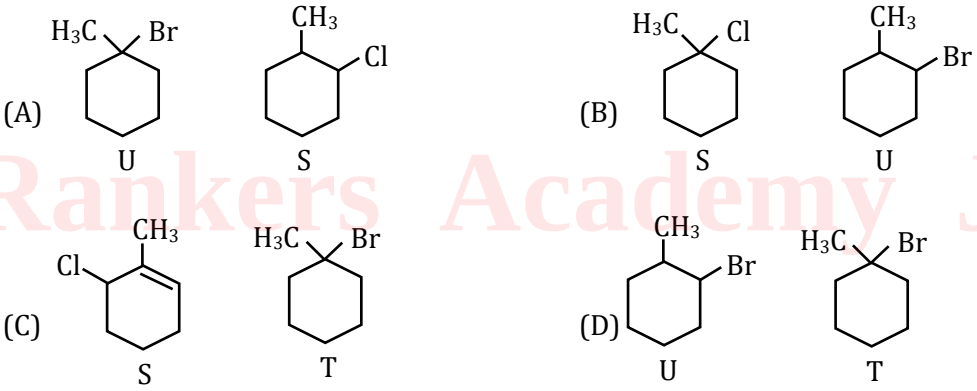
1. An organic compound **P** having molecular formula $C_6H_6O_3$ gives ferric chloride test and does not have intramolecular hydrogen bond. The compound **P** reacts with 3 equivalents of NH_2OH to produce oxime **Q**. Treatment of **P** with excess methyl iodide in the presence of KOH produces compound **R** as the major product. Reaction of **R** with excess *iso*-butylmagnesium bromide followed by treatment with H_3O^+ gives compound **S** as the major product.
The total number of methyl ($-CH_3$) group(s) in compound **S** is _____.

[JEE (Advanced) 2024]

CGR131

2. Choose the correct option(s) for the following set of reactions **[JEE (Advanced) 2019]**

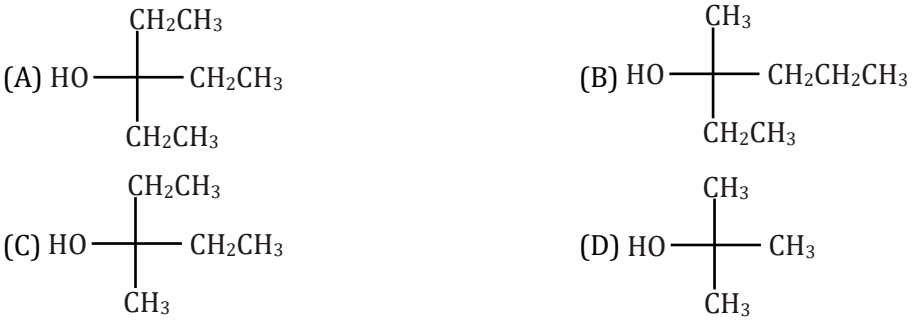




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CGR112

3. CH_3MgBr + Ethyl ester $\rightarrow$ which can be formed as product?
(excess) **[JEE (Advanced) 2003]**



CGR059

4. $(CH_3)_3CMgCl$ reaction with D_2O produces: **[JEE (Advanced) 1997]**

(A) $(CH_3)_3CD$ (B) $(CH_3)_3OD$ (C) $(CD_3)_3CD$ (D) $(CH_3)_3OD$

CGR058

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	C	A	A	C	A	C	D	B
Que.	11	12	13	14						
Ans.	D	C	D	A						

EXERCISE (O-2)

Que.	1	2	3	4	5
Ans.	A,B,C	B,C,D	A,B,C,D	B,D	B,C

EXERCISE (S)

1.	8	**2.**	3	**3.**	0	**4.**	7	**5.**	4
6.	139	**7.**	3	**8.**	9	**9.**	10	**10.**	5
11.	6	**12.**	2.00	**13.**	6.00	**14.**	5.00	**15.**	2.00
16.	3.00								

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	2	1	4	1	3	2	33	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	4	4	2	3	4	3	1	1
Que.	21	22	23	24	25	26	27			
Ans.	2	2	3	4	4	2	1			

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4
Ans.	12,6	B,D	D	A

For More Join: @IITJEE_Advanced